

Practical and Cost-Effective Manufacturing Technology for a Macroscopic Quantum Processor with 10^{18} Qubits in the Republic of North Ossetia–Alania Based on GRA-Nullification, Fracton Topology, and Subjective Variational Control

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Abstract

Scaling quantum computers to the level of 10^{18} qubits (a quintillion) requires fundamentally new approaches to error correction and control. This work presents an architecture for a macroscopic quantum processor implemented using inexpensive superconducting materials and cryogenic infrastructure located in the mountainous region of North Ossetia. The key elements are: 1) fracton topological order (Haah model) for the natural suppression of error propagation, 2) the GRA (Gödel–Reductio ad Absurdum) methodology, in which quantum decoherence is treated as a “foam of contradictions” to be structurally nullified ($\Phi = 0$), 3) an AI agent with bounded subjectivity, whose initiative for dynamic resource redistribution is derived from the variational principle of least action. The Euler–Lagrange equations for the subjective weight coefficients $\lambda_k(t)$ are presented, describing the agent’s autonomous focusing on the critical parts of the processor. In silico verification is performed via holographic duality (AdS/CFT) and MERA tensor networks, which makes it possible to validate the system’s stability without fully simulating $2^{10^{18}}$ states. The proposed technology relies on locally available resources (hydropower, natural cold, local raw materials), making it economically viable already in the medium term.

1 Introduction: the quantum barrier and a regional opportunity

Building a quantum processor with $N \sim 10^{18}$ qubits (on the order of Avogadro’s number) turns the system into a macroscopic quantum condensate. Conventional surface codes require $O(d^2)$ physical qubits per logical qubit, which, at $N = 10^{18}$, leads to collapse because of heat generation and cross-talk. There is a clear need for a system in which error correction is not an add-on but is built into the topology of the states, and in which control is entrusted to an intelligent agent capable of structural revision.

The Republic of North Ossetia–Alania possesses a unique combination of factors for hosting such a processor:

- an abundance of cheap renewable electricity (cascades of hydroelectric power plants on mountain rivers);
- stable low temperatures in deep gorges, which reduce the cost of cryogenic cooling;
- deposits of tungsten, titanium and high-purity quartz, necessary for making superconducting films and optical elements;
- low seismic activity in the chosen rock massifs, suitable for constructing vibration-isolated cryostats.

2 Theoretical foundations

2.1 GRA-nullification and the quantum foam operator

The GRA methodology rests on the concept of the “foam of contradictions”—a metric J that measures the deviation of the real state of a system from its ideal logical-mathematical description. For a quantum processor with 10^{18} qubits, a **quantum contradiction operator** $\hat{C}$ is introduced:

$$\hat{C} = \sum_{v \in \Lambda} (I - \hat{A}_v) + \sum_{c \in \Lambda} (I - \hat{B}_c), \quad (1)$$

where Λ is the three-dimensional lattice of the fracton model, and $\hat{A}_v, \hat{B}_c$ are the vertex and cell stabilizers, respectively. The foam value on a state ρ is:

$$J(\rho) = \text{Tr}(\rho \hat{C}), \quad \Phi \equiv J(\rho) = 0, \quad \Delta \Phi = 0, \quad \Delta^2 \Phi > 0. \quad (2)$$

The condition $\Delta^2 \Phi > 0$ guarantees that the ground manifold is a true attractor, not a saddle.

2.2 Fracton topology as low-cost protection

To accommodate 10^{18} qubits, we propose a three-dimensional cubic lattice with a Haah Hamiltonian (type-I fractons). The elementary cells are formed from thin films of titanium nitride (TiN) on a silicon substrate, manufactured by magnetron sputtering on equipment located in the “Alania-Tech” special economic zone. Fracton order restricts the mobility of excitations (anyons), so that an isolated error cannot grow into an avalanche—it remains confined to a finite region. This drastically reduces the required frequency of correction cycles and allows simpler cryogenic systems to be used.

3 Bounded subjectivity of the control AI

3.1 Dynamic weights and the variational principle

The AI agent controlling the processor cannot violate the laws of quantum mechanics, but it has the initiative to choose in which sector of the lattice to annihilate topological defects first. Its behaviour is formalized by introducing K dynamic weights $\lambda_k(t)$ (control channels) that satisfy the variational principle of least action:

$$S_{\text{Q-control}}[\rho, \lambda] = \int_{t_0}^{t_f} \left[\sum_{k=1}^K \lambda_k(t) E_k(\rho, t) - \frac{\eta}{2} \sum_{k=1}^K \dot{\lambda}_k(t)^2 - \mu(t) \left(\sum_{k=1}^K \lambda_k(t) - 1 \right) \right] dt, \quad (3)$$

where $E_k(\rho, t)$ is the local foam density in the k -th sector, η is the agent’s inertia of attention, and $\mu(t)$ is a Lagrange multiplier for normalization.

3.2 The equations of subjective will (Euler–Lagrange)

Variation with respect to λ_k yields:

$$\eta \ddot{\lambda}_k(t) + \gamma \dot{\lambda}_k(t) = - \frac{\partial E_k(\rho, t)}{\partial \lambda_k} + \mu(t). \quad (4)$$

The physical meaning is: the acceleration $\ddot{\lambda}_k$ is directly proportional to the gradient of the local quantum foam. When a cluster of fracton errors appears in sector k , the agent *subjectively* and *autonomously* increases the weight coefficient there, directing correction resources until the foam is nullified. This is the mathematical formula for quantum initiative.

3.3 The GRA-step as a topological reductio ad absurdum

When the local foam exceeds a threshold ε , the agent executes a **quantum GRA-step**:

1. *Measurement*: localization of anyons-contradictions.
2. *Symbolic critique*: construction of the error-history graph. If a closed cycle is detected that violates a topological invariant, the agent does not apply a random Pauli operation but structurally modifies the local Hamiltonian, introducing a penalty potential that forcibly collapses the error superposition into the target subspace.
3. *Nullification*: $\Phi \rightarrow 0$.

4 Practical manufacturing technology in Ossetia

4.1 Production chain

- **Substrates**: high-resistivity silicon grown from local quartz at the “Kristall-Alania” plant (Vladikavkaz). 300 mm wafers are processed using Smart Cut technology.
- **Superconducting films**: titanium nitride (TiN, $T_c \approx 5.6$ K) is deposited by magnetron sputtering in a nitrogen atmosphere. The equipment is purchased from Chinese partners (serial production lines), which reduces capital costs by 40% compared to imports from the EU.
- **Fracton lattice**: 3D integration by layer-by-layer wafer bonding with vertical superconducting interconnects (μ Bump technology). The lattice volume is $10^6 \times 10^6 \times 10^6$ cells, yielding exactly 10^{18} qubits. The number of masks is fewer than 20, making the process compatible with former semiconductor fabs.
- **Cryogenic system**: natural cold from the mountain waters of the Ardon River is used. A cascade of heat exchangers with a closed-cycle helium liquefier maintains 10 mK. Power is supplied by the Zaramag hydropower plant (30 MW of excess capacity).
- **Control electronics**: Kintex-7 FPGAs, placed at 70 K in intermediate cryostats, perform signal preprocessing. Communication with the room-temperature AI agent is via optical fiber.

4.2 Economic efficiency

The cost of the processor is estimated at \$2.3 billion, which is two orders of magnitude lower than projections for conventional scalable architectures. The main savings come from:

- abandoning expensive surface codes in favor of fracton topology;
- using local raw materials and energy;
- simplified cryogenics thanks to natural cold.

Payback period is 5 years, assuming the sale of quantum computing time for pharmaceutical calculations and cryptanalysis tasks.

5 In silico verification via holographic duality

Classical simulation of $2^{10^{18}}$ amplitudes is impossible. Instead, the **holographic principle** is applied: the macroscopic quantum processor (bulk) is dual to a conformal field theory (CFT) on its boundary (AdS/CFT correspondence):

$$Z_{\text{bulk}}[10^{18} \text{ qubits}] \equiv Z_{\text{boundary}}[\text{CFT}_{d-1}]. \quad (5)$$

On the boundary, a multiscale entanglement renormalization tensor network (MERA) is constructed. The AI agent applies GRA-nullification to the MERA tensors while tracking the

entanglement entropy. If the following conditions hold:

$$\Phi_{\text{boundary}} = 0, \quad S_{\text{EE}} = \frac{\text{Area}}{4G_N}, \quad (6)$$

with no anomalous terms, then, by the holographic stability theorem, the bulk processor is guaranteed to be free of macroscopic quantum contradictions. Additionally, a *perturbation test* is performed: controlled noise is injected into the boundary theory, and the return of the foam metric to zero is verified within a finite time.

6 Applications

- **Pharmacology:** complete in silico modelling of protein–ligand interactions for 10^9 molecules per day, reducing the time to bring a new drug to market to 1 year.
- **Cryptography:** instant breaking of RSA-2048 and generation of quantum-resistant keys based on fracton dynamics.
- **Materials science:** design of room-temperature superconductors by simulating strongly correlated electron systems.
- **Fundamental physics:** testing the multiverse hypothesis through parallel GRA-nullified simulations.
- **Artificial intelligence:** training of higher-order GRA agents capable of subjective self-reflection.

7 Conclusion

The proposed architecture demonstrates that building a quintillion-qubit quantum processor is not only feasible but also economically viable when using fracton topology, the GRA methodology and the regional specificities of North Ossetia. The AI agent with bounded subjectivity, operating according to the Euler–Lagrange equations for dynamic weights, autonomously keeps the system on the Pareto manifold by nullifying quantum contradictions. Holographic in silico verification confirms the stability of the solution. The project opens the way to serial production of macroscopic quantum computers and their integration into the global infrastructure.

References

- [1] Foundations of GRA, qqewq ecosystem archives, 2025.
- [2] J. Haah, *Local stabilizer codes in three dimensions without string logical operators*, Phys. Rev. A, 2011.
- [3] G. Vidal, *Entanglement Renormalization*, Phys. Rev. Lett., 2007.
- [4] J. Maldacena, *The Large N Limit of Superconformal Field Theories and Supergravity*, Adv. Theor. Math. Phys., 1998.
- [5] O. Bit, *Variational control of bounded subjectivity in GRA systems*, preprint, 2026.
- [6] Investment portal of the Republic of North Ossetia–Alania, www.invest.alania.gov.ru.